

Title

Author(s)

Affiliations of author(s)

9:16 ratio of Digital Poster Portrait Template

This template has been designed to show clearly within the digital poster screen and app that will be used for the Conference.

Please clear all text and objects on this template before designing your poster. Your poster should include the following sections: Introduction, Materials & Methods, Results, and Conclusion.

We recommend you use standard system fonts such as Calibri, Verdana, Times New Roman, or Arial.

Recommended font sizes:
48-60pt for Titles/Headings
24-30pt for Text

You are limited to 1 slide within your digital poster.

Format should be submitted in PDF only.

Dosimetric evaluation of an endorectal balloon with radiochromic ...

Young Goh¹, Soonki Min², Se Byeong Lee¹, Dongho Shin¹, Joo-Young Kim¹,
Dae Yong Kim¹ and Young-Kyung Lim¹

*Proton Therapy Center, National Cancer Center, Goyang, Korea ¹,
Department of Radiation Oncology, Chosun University Hospital, Gwangju, Korea ²*

ABSTRACT

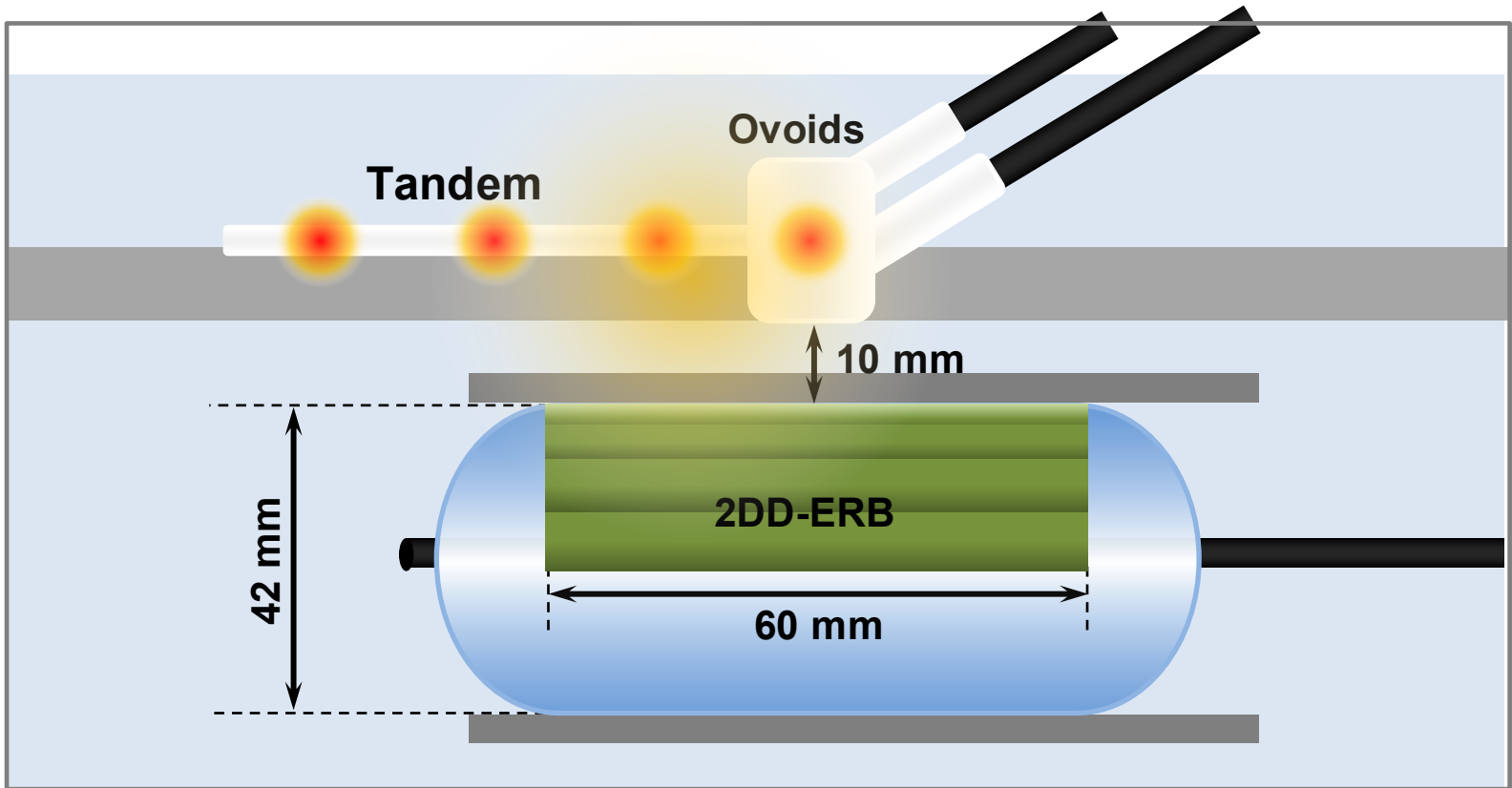
Two-dimensional (2D) in vivo rectal dosimetry using an endorectal balloon (ERB) with an unfoldable

1. Introduction

- HDR brachytherapy for cervical cancer
 - High fraction size, e.g. 5 Gy/fraction or more
 - Steep dose gradient
 - Organs at risk (OAR) toxicity correlated to their volumetric dose, e.g. D(2 cm³)
 - Lack of image-guidance
 - Dose uncertainty due to applicator setup error, rectum and bladder variations, surrounding heterogeneous medium
- ➔ *Two-dimensional in vivo dosimetry having high resolution and Endorectal balloon (ERB) are needed to know the delivered dose accurately and to deliver the dose correctly!*

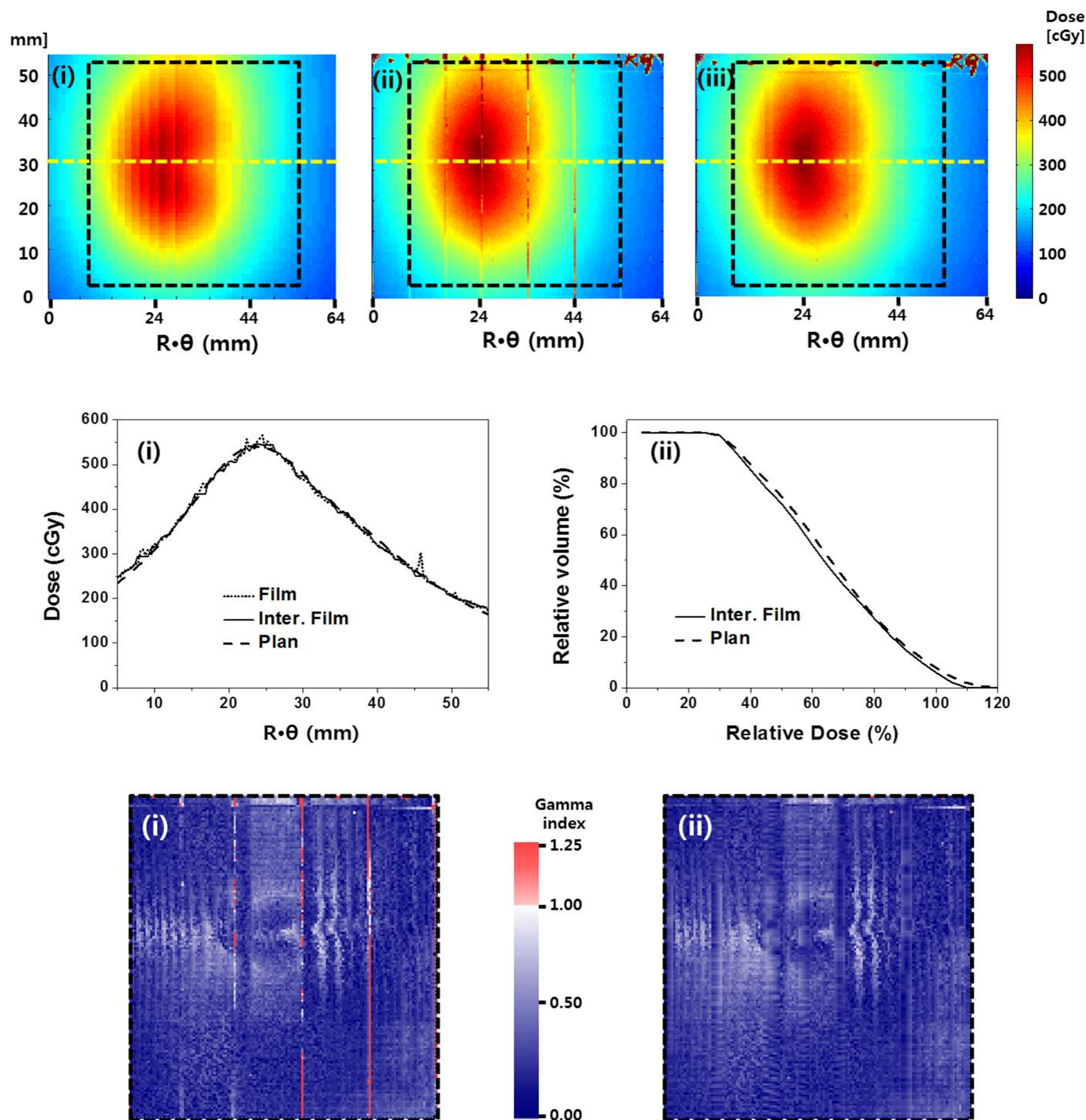
2. Materials & Methods

- Patient selection
 - 20 patients with cervical cancer via MRI-guided intracavitary brachytherapy (ICR).
 - 10 patients for Fletcher (30°, 6 cm tandem, 20 mm ovoids) applicator and 10 patients for Ring (6 cm tandem, 30 mm ring) applicator
 - All patients were prescribed with EBRT (50.4 Gy / 28 fx) + ICR (30 Gy / 6 fx)
- Experimental setup



3. Results

- Dose distributions & Gamma evaluations



4. Discussion

CONCLUSION

The 2D *in vivo* rectal dosimetry using an ERB with an unfoldable EBT3 film is effective and valuable, and will be applicable for HDR brachytherapy for cervical cancer.